Circuit/Schematic Diagram Recognition and Conversion to Machine-Readable Formats: A 2024–2026 Survey with Emphasis on VLMs/MLLMs

TL;DR

- **SPICE netlist is the dominant target**, and the field has converged on a clear verdict: **hybrid pipelines (object detection + explicit wire/net reasoning + LLM serialization/repair) still decisively beat end-to-end MLLMs** for full connectivity. Pipeline systems report 80–97% netlist-level accuracy (Image2Net 80.77% success / 0.116 NED; SINA 96.67%), whereas off-the-shelf MLLMs are strong on component perception but collapse on global connectivity.
- The **perception-vs-connectivity gap is now quantified**: on AMSbench, Gemini-2.5-Pro reaches 94% component-type classification but topology NED remains far from ground truth (0.91) and "connection identification" naming drops to ~0.60; on CircuitSense, Gemini-2.5-Pro's accuracy falls from 80.71% on curated multiple-choice to 19.06% on synthetic equation derivation (~61-point collapse). This is the direct analog of the perception-vs-global-structure gap familiar from table-structure recognition.
- For a Document-AI practitioner, the actionable recipe is: **detector (YOLO/DETR/Swin) for components + designators, deterministic net reconstruction (CCL/Hough + crossing-vs-junction disambiguation), then a fine-tuned code-LLM/VLM (Qwen2.5/3-VL, InternVL) served on vLLM for serialization, error-repair, and bulk corpus/synthetic-data generation** — with graph-isomorphism / netlist-edit-distance validation in the loop.

Key Findings

1. **Output formats.** SPICE netlist is the near-universal target for analog/mixed-signal (AMS) work; graph/adjacency (heterogeneous graphs, pin-level graphs, GraphML/JSON) is the internal representation of choice for GNN methods and validation; **there is no widely adopted "circuit-XML" benchmark standard** — CGHD ships PASCAL-VOC XML for *bounding boxes*, not circuit topology, and EDIF/KiCad/`.asc` formats appear as tool outputs rather than recognition targets. Verilog/RTL is a *digital* target largely disjoint from the schematic-recognition literature (it is generated from specs, not recognized from images). "Compact netlist" DSLs are an emerging token-efficiency play (ChatSCH's Compact Netlist reduces token overhead by ~85%; AnalogGenie's Eulerian pin-sequence).
2. **Three problem regimes with distinct communities.** (a) Printed/typeset AMS schematics from textbooks/papers — the EDA-driven mainstream (AMSNet, Masala-CHAI, Image2Net, Netlistify, SINA, AnalogSeeker). (b) Hand-drawn/handwritten — education/digitization-driven, centered on DFKI's CGHD and the ICDAR/document-analysis community (Digitize-HCD, JUHCCR-v1, Modular Graph Extraction). (c) Board-level/PCB and digital — smaller, newer (SINA's PCB support, ChatSCH/DesignSCH-CoT board-level).

3. **The empirical MLLM result is consistent across benchmarks:** perception (component detection/classification) is largely solved by frontier closed models; connectivity reconstruction and downstream symbolic/design reasoning are not.

Details

1. Output formats and token-length implications

SPICE netlist (dominant). The canonical machine-readable target. Each device line (`M0 1 2 4 4 pch`, `R0 2 0 1k`) binds a component to its nets. This is what AMSNet, Auto-SPICE/Masala-CHAI, Image2Net, Netlistify, SINA, and the AMSbench "perception" task all target. SPICE is attractive because it is directly simulable (ngspice), enabling simulation-equivalence checking and closed-loop repair.

Graph/adjacency representations. Internally, most robust methods reconstruct a graph first and serialize to SPICE second. Variants: (i) component-nodes + net-nodes bipartite/heterogeneous graphs (Image2Net "transforms the circuit netlist into an equivalent heterogeneous graph" for its NED metric; GAT-parsing uses link prediction over detected components); (ii) **pin-level graphs** where each node is a device pin (AnalogGenie), sequentialized into an Eulerian trail for compact autoregressive generation; (iii) subgraph-basis DAGs (CktGNN's two-level nested GNN over a known subgraph basis). JSON/GraphML are used as interchange but no single schema dominates.

XML. Important negative finding for the Document-AI reader: **there is no established circuit-topology-XML recognition benchmark.** CGHD annotations are PASCAL-VOC XML encoding symbol bounding boxes, rotation, and text strings — i.e., detection ground truth, not a netlist. EDIF and KiCad/EESchema/Xschem formats exist as EDA tool I/O (Schemato targets LTspice `.asc`; automatic schematic-generation tools emit Xschem/KiCad), but recognition papers do not evaluate against them as canonical targets. This contrasts sharply with document AI, where formats like ALTO/PAGE-XML and HTML-table markup are standardized recognition targets.

Verilog/RTL. Belongs to the *digital* design-generation literature (VeriGen, AutoChip, RTLLM). It is generated from natural-language specs or synthesized, not recognized from schematic images; the schematic-recognition pipeline and RTL-generation pipeline are largely disjoint communities that share only the "LLM for EDA" umbrella.

Compact-netlist DSLs and token length. Serialized targets have direct VLM-decoding cost. A transistor-level netlist can run to hundreds of lines; Masala-CHAI explicitly analyzes the distribution of "#lines of SPICE code" as a difficulty axis. This motivates: (i) **compact DSLs** — per ChatSCH (Wang et al., ICIC 2026, doi 10.1007/978-981-92-3441-7_32), "we construct DesignSCH-CoT, the first board-level schematic dataset with explicit Chain-of-Thought supervision, comprising 3,781 samples in a Compact Netlist format that reduces token overhead by nearly 85%"; and AnalogGenie's Eulerian pin-sequence "that efficiently describes large circuit graphs"; and (ii) the practical observation that long netlists blow past attention/positional budgets and inflate error rates — the "long-netlist token limit" open problem. For a decoder, every extra net name and designator is more tokens to emit correctly, and a single mis-emitted net silently corrupts topology (exact-match and simulation-equivalence both fail).

2. Approaches

Modular / pipeline methods (current SOTA for connectivity)

The dominant architecture mirrors document structure recognition: **detect primitives → localize connection points → reconstruct adjacency → serialize.** Component detection ≈ cell detection; wire/net reconstruction ≈ table structure/adjacency recovery.

- **AMSNet (arXiv 2405.09045; IEEE LAD 2024; Tao, Shi et al.).** The seminal pipeline+dataset. Component detection (object detection over textbook-harvested symbols) + net detection under two assumptions: nets are solid lines, and intersecting wires without a junction dot are *not* connected. Summary drawn "from 894 schematics"; the initial public release comprised 734 netlists (the baseline Masala-CHAI later cites). Followed by **AMSnet 2.0 (arXiv 2505.09155)** adding AI segmentation for net detection and richer visual info, and aiming at netlist→schematic generation.
- **Parsing Netlists via GAT (Hu, Zhan, Tong; Sensors 2024, 24(1):227, doi 10.3390/s24010227).** Bottom-up graph encoding: an improved **Swin-Transformer** detector with a DC-PAFPN feature-fusion neck, a port-coordinate localization module, and a **Graph Attention Network link-prediction** head. Annotated a 1,200-sample component-detection set and a 3,552-sample circuit set. Canonical example of casting net reconstruction as edge/link prediction over a component graph — directly analogous to adjacency-matrix prediction in table structure recognition.
- **Modular Graph Extraction (Bayer et al., arXiv 2402.11093).** The hand-drawn counterpart, built on CGHD. Explicit modular pipeline: object detection → geometric/structural processing (junctions, wire hops) → graph assembly. Uses drafter splits (e.g., drafters 21–22 for validation) that became a de-facto handwritten baseline.
- **Masala-CHAI (arXiv 2411.14299; evolved from Auto-SPICE).** Fully automated textbook→SPICE pipeline: (i) YOLOv8 schematic/figure extraction from paginated PDFs, (ii) fine-tuned component detectors, (iii) **deep Hough-transform priors for net identification**, (iv) LLM prompt-tuning for netlist repair, (v) post-extraction verification via graph matching. Per Bhandari et al., it curated "a corpus of ~7,500 SPICE netlists with metadata (figure captions, components, and nodes) from 10 textbooks … the largest curated and labeled SPICE dataset available currently," dwarfing the AMSNet baseline of 734 netlists. A later multi-agent variant (Agent-1 lists components, Agent-2 traces connectivity + runs ngspice, a Judge verifies, up to 2 revisions). Earlier **Auto-SPICE (arXiv 2411.14299v1)** produced ~2,100 netlists from a single sample textbook and did the GPT-4o failure-mode case study (NMOS/PMOS confusion, floating nets).
- **Netlistify (MLCAD 2025, doi 10.1109/MLCAD65511.2025.11189145; Huang, Chen, … Po-Hung Lin, Wen-Hao Liu, Haoxing Ren; NVIDIA/NYCU; Best Artifact Award).** Deep-learning-based component recognition + **orientation detection** + learned connectivity extraction (vs fixed geometric rules). Ships a **100,000-image synthetic schematic dataset** for training line detection/interpretation.
- **SINA (arXiv 2607.01609 and DATE'26 version arXiv 2601.22114; Aldowaish et al., Fayazi group).** Fully automated, open-source, **handles both IC- and PCB-level**. YOLOv11 detection + **Connected-Component Labeling (CCL)** for connectivity + OCR for

reference designators + a **VLM (GPT-4o) for designator assignment**, plus a **dedicated crossing-wires detection module** (morphological transforms + contour analysis) to separate junctions from crossings. Validated with **graph isomorphism**. Per Aldowaish et al., "SINA achieves a component detection precision of 98% for IC-level and 94% for PCB-level circuits, along with a text … extraction accuracy of 97.5% … an overall netlist generation accuracy of 96.67%, which is 2.72x higher compared to a state-of-the-art open-source method [Masala-CHAI]," and adds a crossing accuracy of 87% with crossing precision of 100%. The detector is fine-tuned on >700 annotated schematics. (The DATE'26 version reports the alternate figures: 96.47% overall accuracy, detector F1 96.47% with precision 94%/recall 99%/weighted mAP 98%.)

- **Img2Sim / Img2Sim-V2 (Gurbuz, Balta et al.).** CAD tool for user-independent simulation directly from circuit images — an applied pipeline in the same family.
- **Vision-Based Topology-Consistent Structural Parsing (PMC13259090).** Hand-drawn parsing unifying CGHD + Digitize-HCD + new samples into a **57-class** label system (54 component + text/terminal/crossover). Reports **98.03% component-classification accuracy on JUHCCR-v1** (33,329/34,000), beating a 91.15% reference by 6.88 pts, and CGHD mAP@0.5:0.95 under the dataset-paper and Bayer-split protocols.

Validation primitives across these: graph isomorphism (SINA, Masala-CHAI), netlist edit distance (Image2Net), and SPICE-simulation checks (ngspice in the loop).

End-to-end VLM/MLLM methods

Feeding a schematic image to an MLLM and decoding a netlist directly. **Consistent empirical finding: off-the-shelf MLLMs are strong at component perception/classification but weak at full connectivity/netlist reconstruction.**

- **AMSNet GPT-4/GPT-4o experiments** (and the Auto-SPICE/Masala-CHAI case studies) first documented this: GPT-4o mis-lists components, confuses NMOS/PMOS, and produces floating nets.
- **AMSbench (arXiv 2505.24138; Shi et al.).** ~8,000 questions over AMS-Perception (6k), AMS-Analysis (2k), AMS-Design (68), evaluating GPT-4o, Gemini-2.5-Pro, Claude-3.7-Sonnet, Grok-3, Doubao-1.5-vision-pro, Kimi-VL-A3B, Qwen2.5-VL-72B (+ DeepSeek-R1 on TQA). Per Shi et al., "For component type classification, Gemini performs well, reaching 94% accuracy. Among open-source models, Qwen2.5-VL achieves 86%." But interconnect/netlist recognition lags: topology **NED 0.91 (Gemini)**, 1.40 (GPT-4o), 2.38 (Qwen2.5-VL); connection-*judgment* (true/false) ACC reaches ~0.85 for Gemini, yet connection-*identification* (naming connected elements) ACC is only ~0.60 (Gemini), 0.16 (GPT-4o), 0.14 (Qwen2.5-VL). Authors' verdict: "While the models can produce reasonably accurate predictions for local connections, they fall short in reconstructing the complete netlist. Even netlists produced by the best-performing model, Gemini-2.5-Pro, require substantial modifications to align with the ground truth." Root cause: "MLLMs are inaccurate in assigning connection points (i.e. pins and ports) to their parent components." On analysis, the industry-relevant

trade-off ("Engineer") sub-task is a low point — "Even the best-performing model, GPT-4o, only achieves 58% accuracy." On design Pass@k, Grok-3/Claude-3.7 lead (Pass@3 up to 0.65), but no model reliably generates syntactically correct testbenches.

- **CircuitSense (arXiv 2509.22339; Akbari et al.).** 8,006+ problems, six abstraction levels, Perception (890)/Analysis (7,043)/Design (157). Headline: "Closed-source models achieve over 85% accuracy on perception tasks … yet their performance on symbolic derivation and analytical reasoning falls below 19%." Gemini-2.5-Pro scores 100% component detection, 100% connection identification, 95% function classification on perception. Per Akbari et al., "Gemini-2.5-Pro's accuracy drops from 80.71% on curated multiple-choice problems to 70.32% on open-ended versions, and further collapses to 19.06% on synthetic problems requiring direct equation derivation" — an ~61.65-point collapse. Failure-point analysis pins the bottleneck at output-impedance derivation (8%). Design accuracy correlates with symbolic ability (Gemini-2.5-Pro: 36.38% schematic-level, 67.27% block-level, 51.35% hierarchical).
- **Image2Net (arXiv 2508.13157; Xu et al.; ISEDA 2025).** Positions hybrid vs pure-LLM head-to-head and wins: **80.77% success rate, +34.62–45.19 pts over prior works; 0.116 average NED, 62.1–69.6% lower** than SOTA. Releases device-identification, crossing-identification, and device-orientation datasets (2,914 diagrams, 84,195 annotations) plus a 104-pair manually verified eval set and the **NED metric**.

Fine-tuned VLMs and text-side LLMs

- **AnalogSeeker (arXiv 2508.10409).** Fine-tunes **Qwen2.5-32B-Instruct** via a "neighborhood self-constrained SFT" (NSC-SFT) that adds a KL term anchoring to the base model — an explicit **catastrophic-forgetting mitigation.** Per the paper, "we train the Qwen2.5-32B-Instruct model to obtain AnalogSeeker, which achieves 85.04% accuracy on AMSBench-TQA … with a 15.67% point improvement over the original model, and is competitive with mainstream general-purpose LLMs like DeepSeek-v3 and GPT-4o" (base 69.37%; QwQ-32B 81.54%; GPT-4o 73.99%; DeepSeek-v3 84.41%). Finding: at textbook-corpus scale, SFT alone beats CPT+SFT (<1% delta from adding CPT); "Instruct" models are more robust to fine-tuning than "reasoning" models. Note this is a **text-only** model (AMSBench-TQA is text QA), not a VLM.
- **NetlistGPT / ChatSCH (Springer, doi 10.1007/978-981-92-3441-7_32).** Board-level, logic-geometry-decoupled: **NetlistGPT (fine-tuned Qwen2.5-Coder-7B-Instruct)** does component selection + netlist construction; deterministic algorithms do layout + Manhattan routing. Introduces **DesignSCH-CoT** (3,781 Compact-Netlist CoT samples) and **DesignSCH-Bench** (232 expert tasks). Exemplifies the code-LLM route for the serialization stage.
- **LoRA fine-tuning of Qwen2.5-VL / Qwen3-VL / InternVL** on schematic data is the natural open-model recipe; DiagramNet-style progressive training + decoupled multi-agent workflows report large gains (a 3B model + workflow beating GPT-5/Claude-Sonnet-4/Gemini-2.5-Pro by >2× end-to-end on that benchmark, and boosting Gemini-2.5-Pro's Task-1 performance 128.7×) — evidence that domain adaptation + explicit structure closes much of the frontier-model gap.

Agentic / multi-agent frameworks

- **SPARC (arXiv 2606.20643; Sadia et al.).** Multi-agent circuit-diagram QA that grounds reasoning in executable SPICE: a planner decides #simulations, agents synthesize/repair ngspice programs (patch generation + repair), and an analysis stage interprets outputs. **83% accuracy, up to +58 pts over baselines.** (Distinct from the same-acronym VLM test-time-scaling "Separating Perception And Reasoning Circuits," arXiv 2602.06566.)
- **MuaLLM (arXiv 2508.08137; Fayazi group).** Multimodal LLM agent for circuit-design assistance with hybrid contextual RAG.
- **TopoSizing, HeaRT, SABLE.** Netlist-understanding/optimization agents: TopoSizing organizes raw netlists into device-module-stage hierarchies with LLM hypothesis-verification loops feeding Bayesian optimization; HeaRT is a hierarchical circuit-reasoning-tree engine for AMS design optimization. These consume netlists (post-recognition) rather than doing recognition, but define the downstream Pass@k tasks that make recognition worthwhile.

3. Dataset catalog

Dataset	Year / venue	Size	Classes / taxonomy	Output / annotation	Regime	Access	License
AMSNet / AMSNet-1.0 (arXiv 2405.09045)	2024, IEEE LAD	~894 schematics; 734 netlists public	transistor-level (MOS, R, C, junctions)	SPICE netlists + component/net detection	Printed (textbooks)	Partial public; tool+full set promised	⚠ Not clearly stated
AMSnet 2.0 (arXiv 2505.09155)	2025	Larger; AI-segmented nets	AMS incl. richer visual info	Schematic–netlist pairs, segmentation	Printed	Project pages	⚠ Not clearly stated
AMSnet-KG (TODAES 2025, doi 10.1145/3736166; arXiv 2411.13560)	2025, ACM TODAES	Schematics+netlists + knowledge graph	AMS w/ functional/perf annotations	Netlists + KG (RAG for auto-design)	Printed	ACM DL (paper); data via project	⚠ Not clearly stated
AMSbench (arXiv 2505.24138)	2025	~8,000 QA (Perception 6k / Analysis 2k / Design 68)	AMS	MCQ/F1/NED/pass@k; netlist-from-schematic	Printed	HF: wwhhyy/AMSBench (+AMSbench/AMSbench)	⚠ Per HF card (unverified)
Masala-CHAI (arXiv 2411.14299)	2024/25	~7,500 netlists, 10 textbooks	passive+active analog	SPICE + captions/KPIs; graph-match verify	Printed	GitHub jitendra-bhandari/Masala-CHAI	Open-source (per repo LICENSE; SPDX unverified)
Auto-SPICE (arXiv 2411.14299v1)	2024	~2,100 netlists (1 textbook)	analog	SPICE + metadata	Printed	GitHub (same repo)	Open-source (per repo; SPDX unverified)

Dataset	Year / venue	Size	Classes / taxonomy	Output / annotation	Regime	Access	License
Image2Net datasets (arXiv 2508.13157)	2025, ISEDA	2,914 diagrams / 84,195 annotations; 104-pair eval	rich device types + crossings + orientation	device-ID / crossing-ID / orientation + netlists; NED metric	Printed (mixed styles)	Open-sourced (project repo)	⚠ Per repo (unverified)
Netlistify dataset (MLCAD 2025)	2025	100,000 schematic images	AMS components + orientation	line detection / netlist interpretation	Printed (synthetic)	GitHub NYCU-AI-EDA/Netlistify (+ pretrained models)	⚠ Per repo (unverified)
SINA benchmark (arXiv 2607.01609 / 2601.22114)	2026, DATE	>700 annotated (detector); IC+PCB eval	IC + PCB components	SPICE netlists; graph-iso validation	Printed + some hand-drawn	Open-sourced	⚠ Per repo (unverified)
AnalogGenie dataset (arXiv 2503.00205, ICLR'25)	2025	>3,000 distinct topologies (×70 aug); ~2,165 raw netlists	Op-Amp, LDO, bandgap, comparator, PLL, LNA, PA, mixer, VCO …	pin-level graph / Eulerian sequence	N/A (topology, not images)	GitHub xz-group/AnalogGenie; HF JianGao666/AnalogGenie	✅ **MIT** (confirmed)
CktGNN / OCB (arXiv 2308.16406, ICLR'23)	2023	10,000 op-amps (Ckt-Bench-101/301)	op-amp subgraph basis	DAG graphs + specs (igraph/ptgraph/tensor)	N/A (graphs)	GitHub zehao-dong/CktGNN	⚠ Per repo (unverified)
CircuitSense (arXiv 2509.22339)	2025/26	8,006 problems (2,986 curated + 5,020 synthetic); Perception 890 / Analysis 7,043 / Design 157	18 component types, 6 hierarchy levels	symbolic equations (SymPy-checked), perception, design	Printed + synthetic	Project data	⚠ Per release (unverified)

Dataset	Year / venue	Size	Classes / taxonomy	Output / annotation	Regime	Access	License
CGHD (Zenodo 14042961 / TC-11 / Thoma & Bayer, DFKI)	2021→2024, ICDAR	3,173 annotated images, 30 drafters, ~300 circuits; 245,962 bboxes; 284 segmentation maps; 59 classes	59 object classes (symbols, junctions, wire hops, text)	bboxes (PASCAL-VOC XML), rotation/mirror, text strings, segmentation	**Hand-drawn**	Zenodo / TC-11; HF lowercaseonly/cghd; GitHub DFKI/cghd	✅ **CC BY 4.0** (data, confirmed); code MIT
Digitize-HCD (Data in Brief 2025, doi 10.1016/j.dib.2025.111315; Mendeley rngcz5wtv8)	2025	1,277 images, 176 volunteers; 18,602 instances; 11,936 text labels	17 component-symbol classes	symbol detection + text label + **port-location heatmaps** (JSON/PNG)	**Hand-drawn**	Mendeley Data (rngcz5wtv8)	✅ **CC BY 4.0** (Data-in-Brief / Mendeley standard)
JUHCCR-v1 (Sci. Reports 2025, s41598-025-22404-5)	2025	20 component classes + synthetic augmentations (34,000 test instances used downstream)	20 hand-drawn component types	component images + class labels (+ orientation/distortion aug)	**Hand-drawn**	GitHub AyushRoy2001/Circuit-Component-Analysis; Nature	Article **CC BY 4.0**; data/code per repo (unverified)
DesignSCH-CoT / **DesignSCH-Bench** (ChatSCH, Springer)	2025/26	3,781 CoT samples; 232 eval tasks	board-level components	Compact-Netlist + Chain-of-Thought	Board-level (printed)	Springer (paper)	⚠ Per paper (availability on request)
SpiceNetlist	—	SPICE netlist corpus (text)	analog devices	SPICE text	N/A	(text corpus)	⚠ Not stated

License-column key. ✅ = license verified at source (Zenodo record, HF dataset card, or Data-in-Brief venue policy). ⚠ = an explicit SPDX license was **not** confirmed from the source; the repo/card may still carry one — check the `LICENSE` file before redistribution or commercial

use. “Per repo (unverified)” means the code/data is publicly downloadable but I could not confirm the specific license text. Confirmed permissive: **AnalogGenie (MIT)**. Confirmed share-with-attribution: **CGHD (CC BY 4.0)**, **Digitize-HCD (CC BY 4.0)**, and the **JUHCCR-v1 article (CC BY 4.0)** — note the article license does not automatically cover the linked GitHub data/code. For any dataset marked ⚠, do not assume open/commercial terms; several AMSNet-family releases in particular state no license and are best treated as research-only until clarified with the authors.

Additional 2025–2026 datasets/benchmarks surfaced: **OmniSch** (LMM schematic understanding + spatial netlist graph construction benchmark), **DiagramNet** (10,977 connection annotations + 15,515 CoT QA over system-level diagrams), and the **2024/2025 China Postgraduate IC Innovation Competition · EDA Elite Challenge** contest data (edachallenge.cn) used as a de-facto competitive benchmark.

4. Benchmarks & metrics

- **Component detection:** mAP / mAP@0.5:0.95 (CGHD task, GAT-parsing, Vision-Based Structural Parsing), precision/recall (SINA 98% IC / 94% PCB).
- **Text/designator:** OCR accuracy (SINA 97.5%).
- **Connectivity/net accuracy:** connection judgment ACC + connection identification F1 (AMSbench); link-prediction accuracy (GAT-parsing).
- **Graph-level: graph isomorphism** pass (SINA, Masala-CHAI), **graph edit distance**; **Netlist Edit Distance (NED)** — Image2Net’s contribution, computed on the heterogeneous graph, and adopted as “Topology NED” in AMSbench.
- **Netlist exact-match / success rate:** Image2Net 80.77% success; SINA 96.67%.
- **SPICE-simulation equivalence:** ngspice execution in Masala-CHAI, SPARC.
- **Downstream Pass@k:** Masala-CHAI (fine-tuned models >40% Pass@1 on complex circuits where base ≈0); AMSbench design Pass@3/5/10; CircuitSense design accuracy.

The perception-vs-connectivity gap, quantified. AMSbench: 94% component-type classification vs 0.91 topology NED and ~0.60 connection-identification (Gemini-2.5-Pro), collapsing to 0.14–0.16 for GPT-4o/Qwen2.5-VL. CircuitSense: >85% perception vs 19.06% synthetic equation derivation (80.71→19.06, ~61-point drop for the best model). This is the exact structural signature the Document-AI reader knows from table structure recognition: local cell/component detection is easy; global adjacency/reading-order/net reconstruction is where systems fail, and it is where explicit structural modeling (GNN link prediction, CCL, graph-iso validation) outperforms end-to-end decoding.

5. Parallels to document structure recognition

- **Component detection ≈ cell/word detection.** Same detectors (YOLO, Faster R-CNN, DETR-family, Swin), same long-tail-class and small-object problems; textbook uniformity is exploited exactly as in born-digital document harvesting.
- **Net reconstruction ≈ table structure / adjacency recovery.** Predicting which pins share a net is isomorphic to predicting which cells share a row/column — hence GAT link-prediction and heterogeneous-graph formulations. Endpoint snapping / junction detection ≈ ruling-line and cell-boundary reconstruction.
- **Crossing-vs-junction disambiguation ≈ spanning-cell / merged-cell ambiguity** — a local visual cue (junction dot vs hop) with global topological consequences, which is why SINA and Image2Net build dedicated crossing modules (SINA reports 87% crossing accuracy / 100% crossing precision).
- **The perception-vs-global-structure gap** is the shared failure mode; the field's response (hybrid perception + explicit structure + LLM repair) mirrors the trajectory of table recognition away from pure end-to-end image-to-markup.
- **Synthetic data generation** — rendering schematics from SPICE (Netlistify's 100k, CircuitSense's grid-based generator with SymPy ground-truth) is the direct analog of programmatic table/document synthesis.

Recommendations

Staged build recipe (highest expected accuracy today):

1. **Perception stage.** Fine-tune a strong detector (YOLOv11 as in SINA, or a DETR/Swin variant per GAT-parsing) on CGHD + Digitize-HCD + Image2Net device sets for components, junctions, wire-hops/crossings, and text regions. Target mAP/precision >95% on printed, and expect a domain gap on hand-drawn — train per-regime or mix. Benchmark: if component precision <90%, connectivity cannot recover; fix perception first.
2. **Connection points + net reconstruction.** Add explicit port/pin localization (port-heatmaps as in Digitize-HCD; port-coordinate module as in GAT-parsing) and reconstruct nets deterministically via CCL + Hough with **explicit crossing-vs-junction disambiguation** (morphological/contour, SINA-style). This is the single highest-leverage stage — it is where end-to-end MLLMs fail.
3. **Serialization + repair.** Use a fine-tuned **code-LLM (Qwen2.5-Coder-7B/32B, cf. NetlistGPT) or VLM (Qwen2.5-VL / Qwen3-VL, InternVL)** to assign reference designators/values and emit SPICE, with the schematic image as visual context (SINA-style VLM designator assignment). Prefer a **compact-netlist DSL** intermediate (ChatSCH-style, ~85% token reduction) for long circuits, then expand to full SPICE.
4. **Validation loop.** Verify with **graph isomorphism** against the reconstructed graph and, where component values exist, **ngspice simulation-equivalence** (SPARC/Masala-CHAI closed loop). Use **NED** (Image2Net) as the continuous training/eval signal rather than brittle exact-match.

When to consider pure end-to-end VLM: only for perception-heavy sub-tasks (component listing/classification, captioning, QA) or as a repair/serialization module — not for standalone full-netlist extraction. The benchmarks are unambiguous that this underperforms hybrids today.

Serving with vLLM for bulk conversion + synthetic-data generation:

- Serve Qwen2.5-VL/Qwen3-VL or InternVL with `vllm serve`, using tensor/data parallelism (`--tensor-parallel-size`, `--data-parallel-size`), `--limit-mm-per-prompt` to cap image inputs, `--max-model-len` sized to your longest netlist (native 128K; 64K typical), and `--mm-encoder-tp-mode data` to replicate the small ViT encoder (ViT ≈ 675M vs 72B LM). Use `--gpu-memory-utilization 0.95` to maximize KV cache; on A100-80GB, TP≥2 for the 72B to avoid OOM.
- Exploit vLLM **guided/structured decoding (guided_json / grammar)** to constrain output to a netlist grammar — this directly attacks format-invalid outputs (the open-source-model failure noted in AMSbench, where some models "fail to produce outputs in the correct format") and reduces repair cycles.
- For **synthetic-data generation at scale**, render schematics from a SPICE/topology corpus (AnalogGenie topologies, OCB, CktGNN generators; Netlistify's 100k pipeline; CircuitSense's grid generator with SymPy ground-truth) and use the vLLM-served VLM for augmentation/captioning. Textbook harvesting (AMSNet/Masala-CHAI YOLO-on-PDF) remains the best source of realistic printed styles.

Open-model choices. Qwen3-VL (multiple sizes incl. 235B-A22B MoE) and InternVL3 for perception; Qwen2.5-Coder-7B/32B for serialization/repair; 3–7B fine-tunes are sufficient when paired with an explicit structural stage (DiagramNet-style 3B beating frontier models end-to-end with the right workflow).

Metric thresholds that should change your plan: if topology NED plateaus >0.3 or graph-iso pass-rate <80% after detector improvements, the bottleneck is net reconstruction/crossing disambiguation, not perception — invest there. If designator/value errors dominate, add OCR + VLM cross-checking (SINA). If long circuits fail disproportionately, move to a compact DSL and chunk.

Caveats

- **Reported numbers are authors' own, mostly on self-constructed benchmarks and largely unaudited** (AMSbench, CircuitSense, SINA, Image2Net are arXiv preprints; cross-comparability is limited because NED, success-rate, and iso-pass are defined per-paper). SINA's two versions report 96.67% vs 96.47% overall accuracy — minor version drift.
- **CircuitSense preprint has internal naming inconsistencies** (Gemini-2.0 vs 2.5-Pro; Claude-Sonnet-3.5 vs 4; a Table-7 percentage mismatch on total-impedance vs transfer-function-simplification); treat the best model as Gemini-2.5-Pro per its appendix version spec (`gemini-2.5-pro-preview-0605`).

- **arXiv IDs with "26xx" prefixes** (SINA 2607.01609, SPARC 2606.20643, AnalogRetriever 2604.x, AnalogToBi 2603.x, DATE'26 2601.22114) correspond to 2026 submissions consistent with the current date; several are very recent and may revise.
- **No standardized circuit-XML recognition target exists** — do not expect an EDIF/KiCad benchmark; SPICE + graph are the real targets.
- **Hand-drawn ↔ printed ↔ PCB generalization is unsolved**; models trained on one regime degrade on others (the 57-class unification in Vision-Based Structural Parsing is a partial answer).
- **Catastrophic forgetting** during domain fine-tuning is real; AnalogSeeker's NSC-SFT (KL-anchored SFT) and preference for Instruct-over-reasoning bases are the current mitigations.
- **AnalogSeeker is text-only** (AMSBench-TQA is textual QA); its 85.04% is not a schematic-image result and should not be read as image-netlist accuracy.
- Several "systems" (SpiceNetlist) are thinly documented in public sources; treat as corpora rather than benchmarked pipelines.